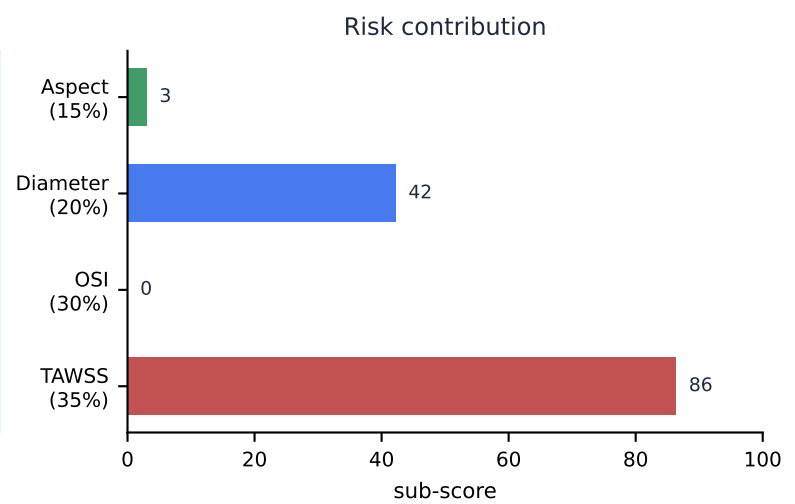
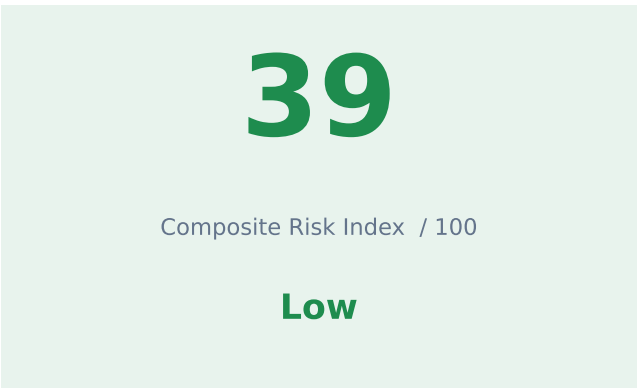


# NeuroFlow CFD Analyst

## Clinical Hemodynamic Assessment

Case PT-2026-0103 · generated 2026-08-04 18:02



### Hemodynamic parameters

| Parameter                    | Value                  | Threshold | Status  |
|------------------------------|------------------------|-----------|---------|
| TAWSS — aneurysm dome        | 0.334 Pa               | < 0.4 Pa  | FLAGGED |
| TAWSS — parent artery        | 2.973 Pa               | reference | normal  |
| OSI — aneurysm dome          | 0.000                  | > 0.2     | normal  |
| RRT — aneurysm dome          | 7.41 Pa <sup>-1</sup>  | > 3.0     | FLAGGED |
| ECAP — aneurysm dome         | 0.000 Pa <sup>-1</sup> | > 1.0     | normal  |
| NWSS (sac / parent)          | 0.112                  | —         | normal  |
| LSAR (relative, <10% parent) | 72.5 %                 | —         | normal  |
| LSAR (absolute, <0.4 Pa)     | 78.6 %                 | —         | normal  |

### Morphology (measured from the reconstructed surface)

|                      |                      |                |              |
|----------------------|----------------------|----------------|--------------|
| Max diameter         | Neck diameter        | Aspect ratio   | Dome-to-neck |
| 5.38 mm              | 4.83 mm              | 0.75           | 1.11         |
| Volume               | Surface area         | Non-sphericity |              |
| 63.4 mm <sup>3</sup> | 61.1 mm <sup>2</sup> | 0.001          |              |

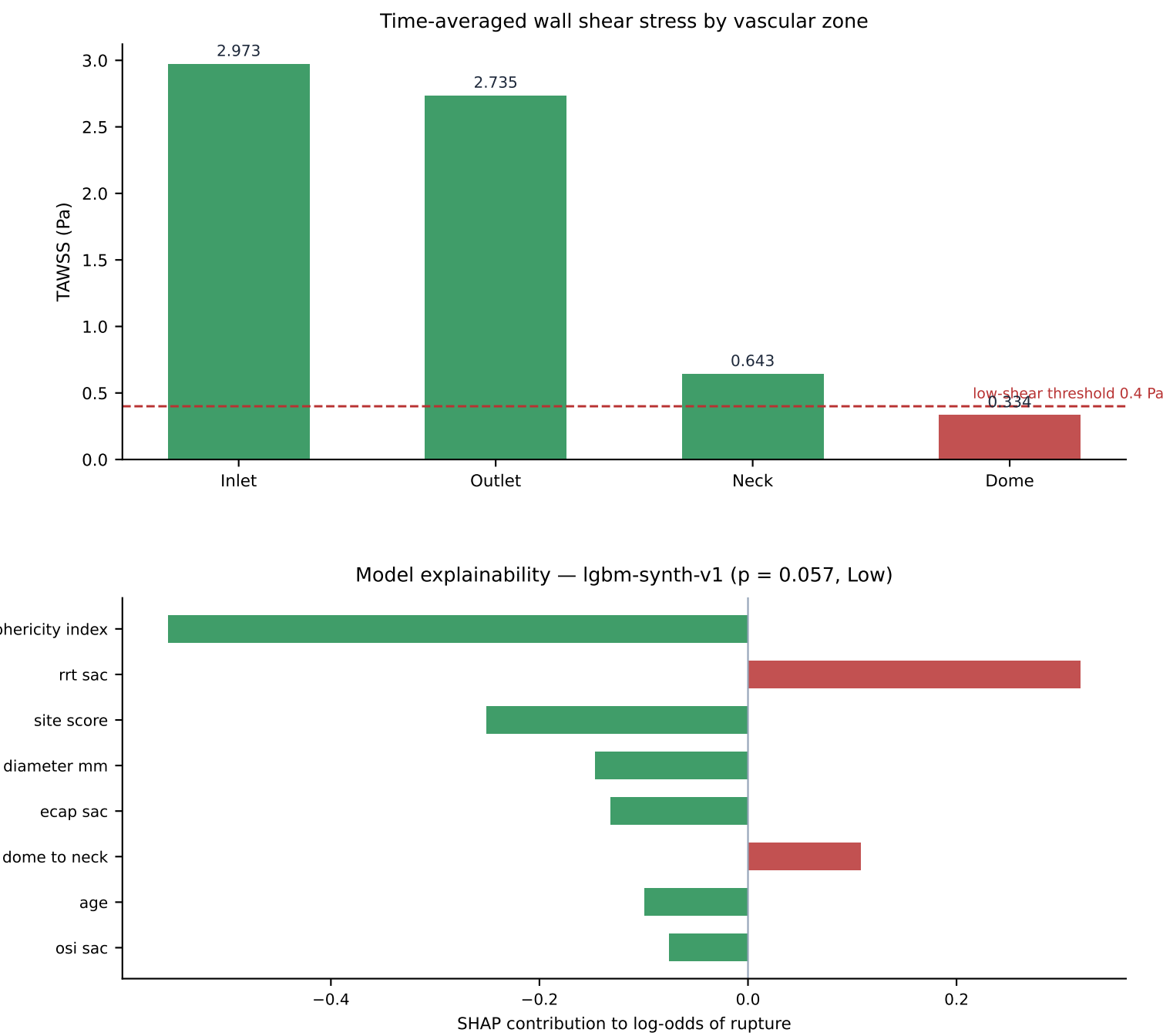
Provenance

Solver: OpenFOAM ESI v2412 · Mesh: cells: 238994 · SIMPLE solution converged in 185 iterations  
WSS converted from OpenFOAM kinematic units (m<sup>2</sup>/s<sup>2</sup>) to Pascals by multiplying by rho = 1060.0. Validated against the analytic Poiseuille solution  $\tau = 4 \cdot \mu \cdot Q / (\pi \cdot r^3)$ .

# NeuroFlow CFD Analyst

## Zone Analysis and Model Explainability

Case PT-2026-0103



### Assessment

Computational fluid dynamics analysis of case PT-2026-0103. Time-averaged wall shear stress at the aneurysm dome measured 0.334 Pa against 2.973 Pa in the parent artery (normalised WSS 0.112), with an oscillatory shear index of 0.000. 72.4% of the sac surface lies below 10% of the parent-artery shear (LSAR). Relative residence time at the dome is 7.41 Pa<sup>-1</sup>. Sac morphology measured from the reconstructed surface: maximum diameter 5.38 mm, neck 4.83 mm, aspect ratio 0.75, volume 63.4 mm<sup>3</sup>. Composite Risk Index 39/100 (Low).